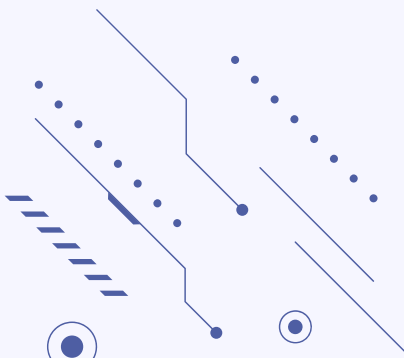
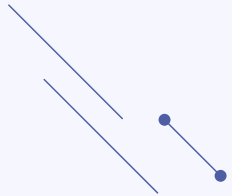
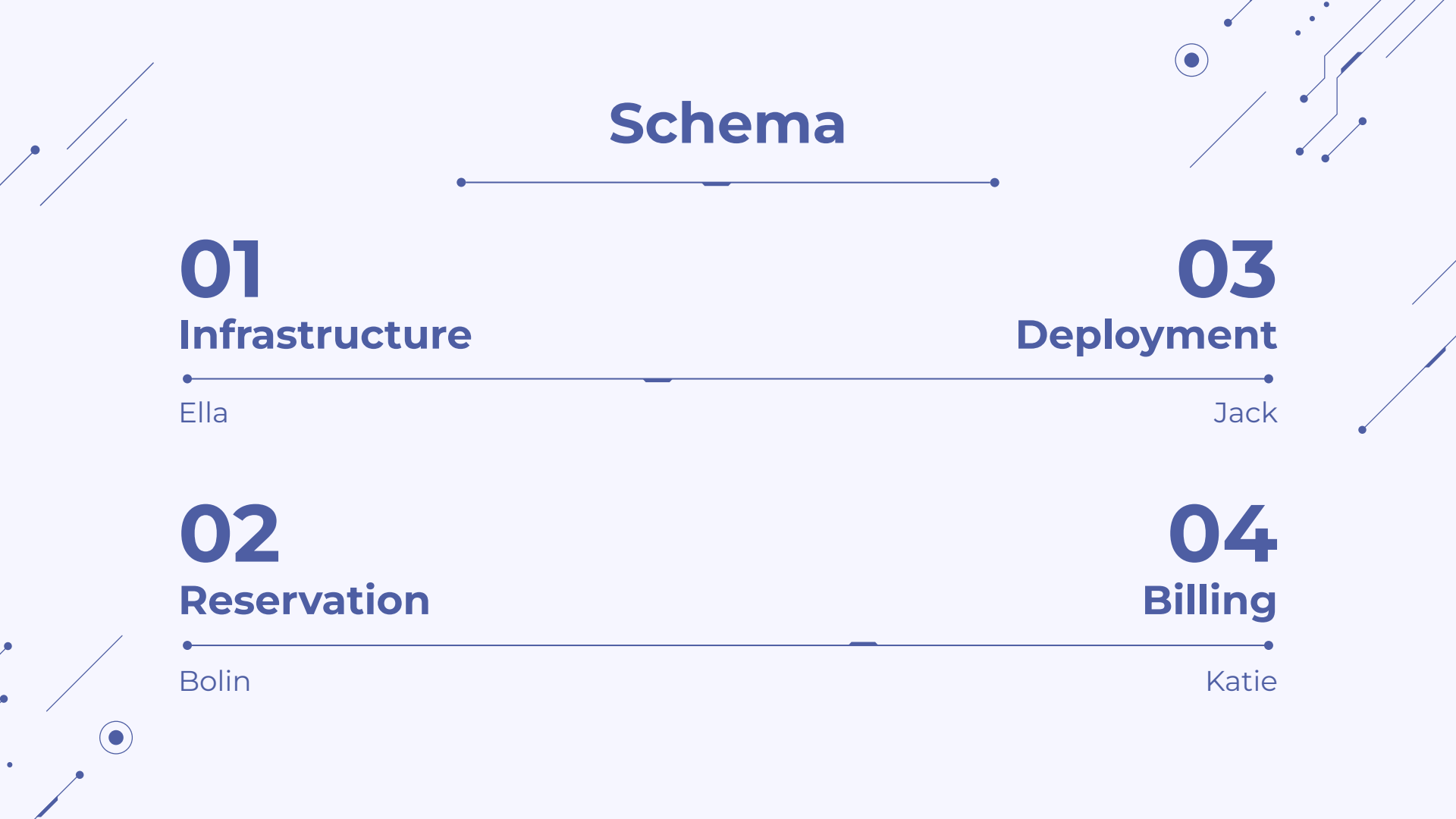


Group 8: Cloudfine Systems

Ella Zhang, Jack Liu, Bolin Yang, Katie Shen



Schema



```
graph TD; S[Schema] --- I[01 Infrastructure]; S --- R[02 Reservation]; S --- D[03 Deployment]; S --- B[04 Billing]; I --- R; I --- D; R --- B; D --- B;
```

01

Infrastructure

Ella

03

Deployment

Jack

02

Reservation

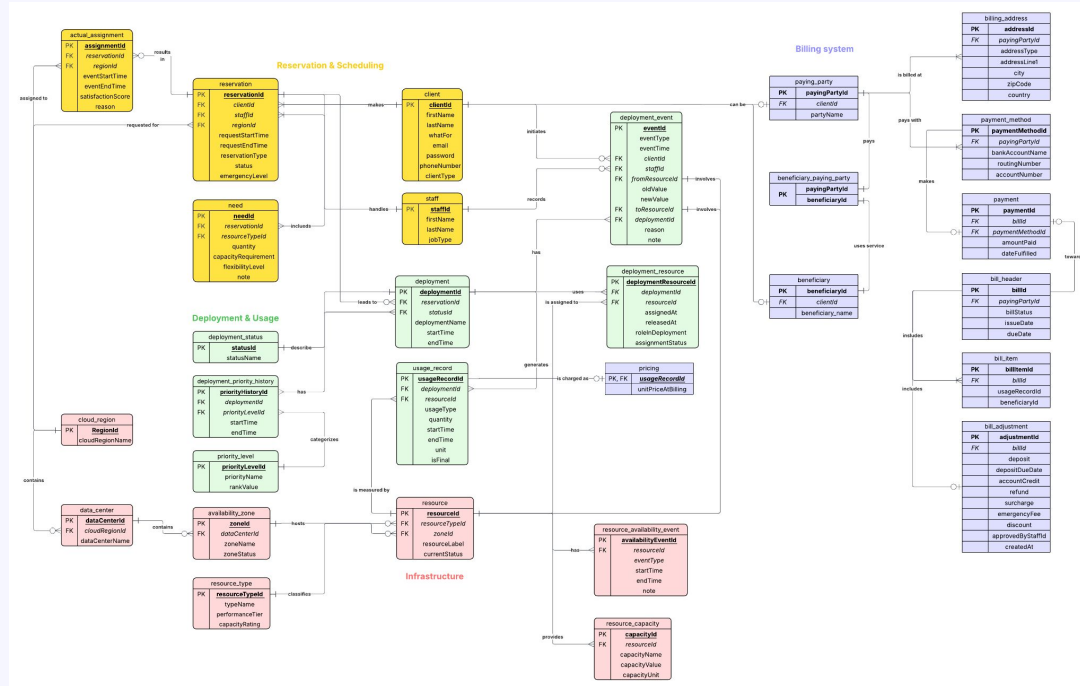
Bolin

04

Billing

Katie

[Link to ERD](#)

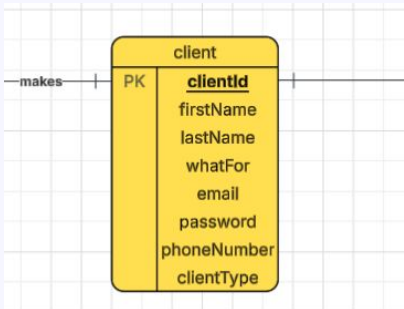


Queries

We designed our queries to answer key operational and analytical questions in the system

Resource Usage	What resources are used by a deployment?	Understand resource allocation
Billing	Who is paying on behalf of who?	Track financial responsibility
Reservation	What time did the client request for reservation?	Analyze client demand
Deployment	How are deployments performing over time?	Monitor system activity
Registration	How do clients join our system?	Expand system services

Schema Design



```
CREATE TABLE client (  
  clientId INTEGER PRIMARY KEY,  
  firstName TEXT NOT NULL,  
  lastName TEXT NOT NULL,  
  whatFor TEXT,  
  email TEXT NOT NULL UNIQUE,  
  password TEXT NOT NULL,  
  phoneNumber TEXT,  
  clientType TEXT NOT NULL CHECK (clientType IN ('individual', 'group', 'organization'))  
);
```

Create all table from ERD to Schema.sql

ERD

Database

Made up some data

```
-- Client & Staff  
7 INSERT INTO client (  
8   clientId,  
9   firstName,  
10  lastName,  
11  whatFor,  
12  email,  
13  password,  
14  phoneNumber,  
15  clientType  
16 ) VALUES  
17 (1, 'Alice', 'Chen', 'analytics workloads', 'alice.chen@cloudnine.demo', 'alice123', '6461234567', 'individual'),  
18 (2, 'Brian', 'Patel', 'backup storage project', 'brian.patel@cloudnine.demo', 'brian456', '3109876543', 'group'),  
19 (3, 'GreenLeaf', 'Labs', 'internal testing and monitoring', 'ops@greenleaflabs.demo', 'green789', '2015551003', 'organization'),  
20 (4, 'Sofia', 'Rivera', 'realtime app hosting', 'sofia.rivera@cloudnine.demo', 'sofia321', '2125552020', 'individual');  
21
```

SQL Query Implementation

```
SELECT d.deploymentId, d.deploymentName,  
       ds.statusName,  
       c.firstName || ' ' || c.lastName AS clientName  
FROM deployment d  
JOIN deployment_status ds  
    ON d.statusId = ds.statusId  
LEFT JOIN reservation r  
    ON d.reservationId = r.reservationId  
LEFT JOIN client c  
    ON r.clientId = c.clientId  
WHERE d.deploymentId = ?;
```

Combine deployment,
status, and client data



Use JOIN to connect related tables



Generate a complete deployment view

SQL Query Implementation

```
SELECT d.deploymentId,  
       COUNT(r.resourceId) AS resourceCount  
FROM deployment d  
JOIN deployment_resource dr  
     ON d.deploymentId = dr.deploymentId  
JOIN resource r  
     ON dr.resourceId = r.resourceId  
GROUP BY d.deploymentId;
```

Count+Group By

SQL to Flask to HTML

The fundamental of the website

- app.py
- Routes to html
- Connects the db

```
{% if reservation %}
<section>
  <h3>Reservation #{{ reservation.reservationId }}</h3>

  <table border="1">
    <tr><th>Client</th><td>{{ reservation.clientName }}</td></tr>
    <tr><th>Client Email</th><td>{{ reservation.email }}</td></tr>
    <tr><th>Staff Handler</th><td>{{ reservation.staffName }}</td></tr>
    <tr><th>Region</th><td>{{ reservation.cloudRegionName }}</td></tr>
    <tr><th>Requested Start</th><td>{{ reservation.requestStartTime }}</td></tr>
    <tr><th>Requested End</th><td>{{ reservation.requestEndTime }}</td></tr>
    <tr><th>Reservation Type</th><td>{{ reservation.reservationType }}</td></tr>
    <tr><th>Status</th><td>{{ reservation.status }}</td></tr>
    <tr><th>Emergency Level</th><td>{{ reservation.emergencyLevel }}</td></tr>
  </table>
</section>
```

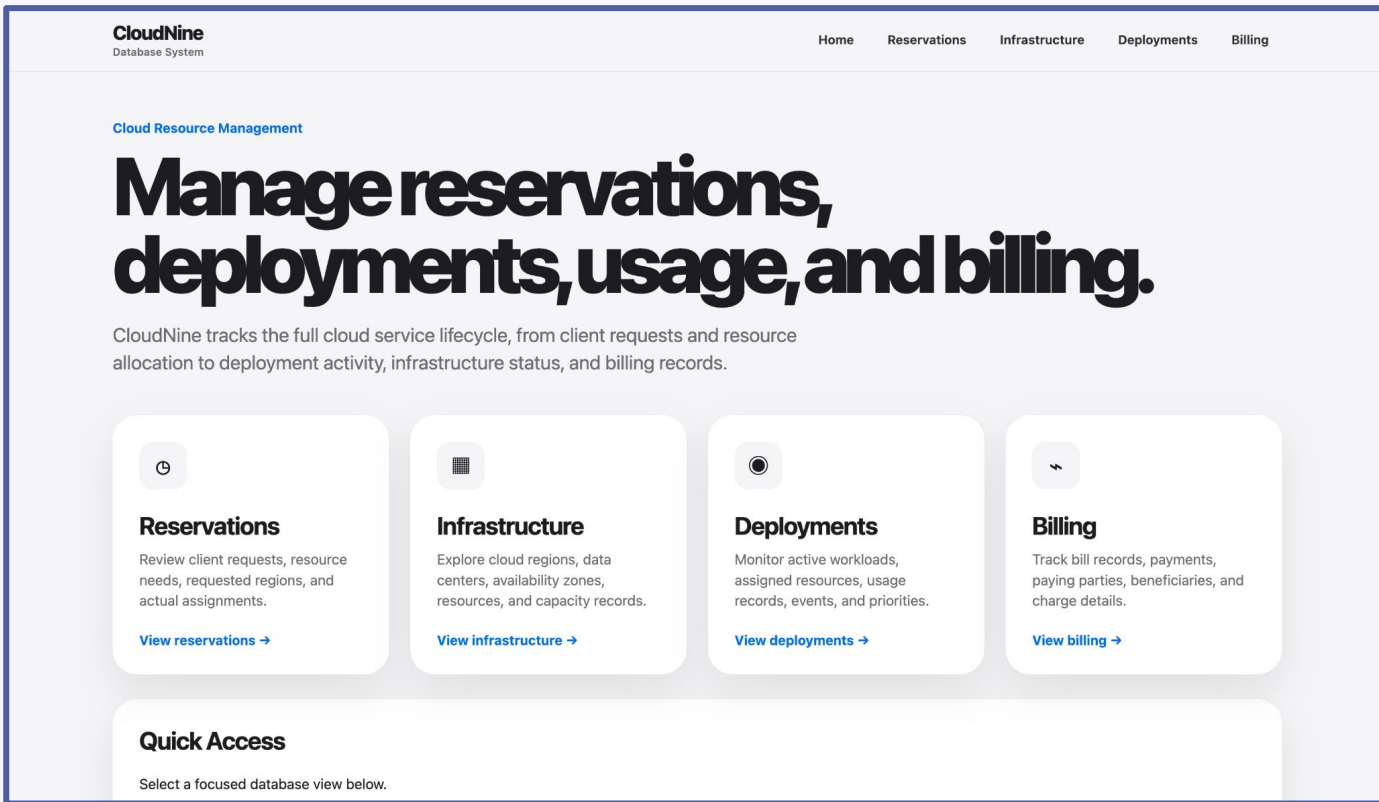
```
@app.route("/clients")
def client_list():
    db = get_db()

    clients = db.execute("""
        SELECT
            clientId,
            firstName || ' ' || lastName AS clientName,
            clientType,
            whatFor,
            email,
            phoneNumber
        FROM client
        ORDER BY lastName, firstName;
    """).fetchall()

    db.close()

    return render_template(
        "clients.html",
        clients=clients
    )
```


Website



Contribution



Thank you